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Clustering biological data

Week 9: Finding groups, and the much harder question of whether they are real.

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By the end of this week you should be able to

Learning outcomes served: LO1, LO2, LO4, LO5

- Apply k-means, DBSCAN and agglomerative clustering, and state the assumptions each makes.
- Select the number of clusters using the elbow method and silhouette analysis, and explain the limits of both.
- Read a dendrogram and choose a cut height defensibly.
- Validate a clustering: stability under resampling, external validation, and biological coherence.
- Explain why a computational cluster is not by itself a biological discovery.

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The biological problem

Part 1 · Why anyone cares about this week's method

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Biological context

- Unsupervised analysis underlies a great deal of modern biology: molecular subtypes of tumours, cell types in single-cell data, community structure in the microbiome, dietary patterns in nutritional epidemiology. Several of these have genuinely changed clinical practice.

- Others turned out to be artefacts of batch, of preprocessing, or of the number the analyst chose for k. Telling the two apart is the skill this week is about.

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The method

Part 2 · Concepts, then the mathematics

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k-means

Alternates assignment and centroid update to minimise within-cluster sum of squares. It assumes roughly spherical, similarly sized clusters, requires k in advance, and converges to a local optimum, so initialisation matters. k-means++ improves the initialisation.

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DBSCAN

Density-based: finds arbitrarily shaped clusters, decides the number of clusters itself, and labels low-density points as noise. In exchange you must choose eps and min_samples, and it degrades in high dimensions where density estimates become unreliable.

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Key concepts

Hierarchical clustering Builds a nested tree of merges. The dendrogram shows structure at every scale at once, which is often more informative for biological data than a single flat partition.

Linkage Single linkage chains and produces straggly clusters; complete linkage produces compact ones; Ward minimises the increase in within-cluster variance and behaves well on roughly spherical data.

Silhouette score For each point, how much closer it is to its own cluster than to the nearest other cluster, from -1 to 1. On standardised WDBC, k-means silhouette is highest at k = 2 (0.343) and declines steadily.

The elbow method Plot within-cluster sum of squares against k and look for a bend. It is a heuristic, it is frequently ambiguous, and it should never be the only justification.

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Key concepts (continued)

Cluster validation Does the partition survive resampling? Does it correspond to something measured independently, such as survival or treatment response? Is it stable under a different algorithm?

Batch effects Samples processed on different days or instruments cluster by batch. This produces beautiful, highly significant, entirely technical clusters, and it is the single most common way unsupervised biological analysis goes wrong.

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k-means and its objective

$$J = \sum \| x_i - \mu(c(i)) \|^2$$

Alternate assignment and centroid update to minimise the within-cluster sum of squares. Guaranteed to converge, but only to a local optimum, so initialisation matters and k-means++ exists to improve it.

$\mu(c)$ the centroid of cluster c

$c(i)$ the cluster assigned to point i

J the distortion, also called inertia

assumes roughly spherical, similarly sized clusters, and k given in advance

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Silhouette coefficient

$$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$$

How much closer a point is to its own cluster than to the nearest other one, from −1 to 1. The mean over all points scores the whole partition.

- **a(i)** mean distance from i to the other points in its cluster
- **b(i)** mean distance from i to the points of the nearest other cluster
- **s ≈ 1** well clustered; s ≈ 0 on a boundary; s < 0 probably in the wrong cluster

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Choosing k on the real WDBC data, measured

Two clusters is the best-supported choice, and it corresponds to the benign/malignant labels. Note that 0.343 is a moderate score, not a strong one: the two groups overlap.

k	Inertia	Silhouette
2	11,595	0.343
3	9,972	0.314
4	9,072	0.283
5	8,406	0.158
6	7,880	0.160
8	7,072	0.147

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Three algorithms, three sets of assumptions

Choosing an algorithm is choosing a set of assumptions about the shape of the answer.

	k-means	DBSCAN	Agglomerative
Number of clusters	You choose k	Found automatically	Cut the dendrogram
Cluster shape	Roughly spherical	Arbitrary	Depends on linkage
Can refuse a point?	No	Yes, as noise	No
Main parameter	k	eps, minPts	linkage, cut height
Degrades when	Clusters are elongated	Dimension is high	n is large: $O(n^2)$ or worse

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A computational cluster is not a biological discovery

When cluster membership aligns with a technical variable, the technical explanation is the default and the biological one carries the burden of proof.

- k-means cannot decline. Ask for four clusters and you get four, whatever the data
- Validate by stability: bootstrap, recluster, and measure how often pairs stay together
- Validate externally: does the partition correspond to something measured independently, such as survival or treatment response?
- Validate across algorithms: does a different method with different assumptions agree?
- Check batch structure FIRST. Samples processed on different days or instruments cluster by batch
- A batch effect produces beautiful, highly significant, entirely technical clusters, and no clustering metric detects it

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Worked example

Part 3 · k-means will always give you k clusters

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k-means will always give you k clusters

- Run k-means with k = 4 on the standardised WDBC data and you get four clusters, each with a centroid, each with members. The algorithm cannot decline. It has no mechanism for reporting that the data does not contain four groups.
- The silhouette scores measured on this data tell the more useful story: 0.343 at k = 2, 0.314 at k = 3, 0.283 at k = 4, and then a marked drop to 0.158 at k = 5. Two clusters is the best-supported choice, and it corresponds to the benign and malignant labels, which is reassuring.
- But 0.343 is a moderate score, not a strong one. The two groups overlap.
- Now consider what happens if you had not had labels. You would have found two clusters, and you would have been right. If you had instead reported five clusters as five novel subtypes, the algorithm would have supported you just as obediently.
- The discipline has to come from the validation, not from the clustering.

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The algorithm answers the question you asked. It cannot tell you the question was wrong.

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Try this interactively

Clustering Simulator

Is this cluster structure biological, or is it the batch?

`weeks/week09.html` · `Interactive exploration`

Open the week page on the course website and scroll to Interactive exploration.

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The foundation-model connection

Part 4 · Clustering in embedding space

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Clustering in embedding space

- Clustering foundation-model embeddings is increasingly common and the caveats compound rather than cancel. Embedding spaces are anisotropic, so Euclidean distance may not behave as the algorithm assumes and cosine distance is often more appropriate.
- The clusters reflect what the pretraining corpus made salient, which may be taxonomy or sequence length rather than the function you care about. And the model has no notion of your batch structure, so it will not protect you from batch effects.
- Clusters in embedding space are hypotheses, at exactly the same evidential level as clusters in any other space.

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Interpretation and practical

Part 5 · What the result means, and what you will do with it

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You cluster 200 tumour samples and find three groups. Samples were collected at three hospitals. Design the single analysis that would most efficiently tell you whether you have found biology or logistics.

Think like a scientist. Bring your answer to the laboratory.

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Clusters, and whether to believe them

This week's laboratory · Cluster real data, choose k defensibly, and then try seriously to disprove your own result. · 2 hours

- Standardise WDBC and run k-means for k = 2 to 10. Plot inertia (the elbow) and silhouette against k. Choose k and justify it in one sentence.
- Compare your clusters with the true diagnosis using the adjusted Rand index. Note that in a real discovery setting you would not have this label.
- Run DBSCAN. Tune eps using a k-distance plot rather than by trial and error. Report how many points it labels as noise, and inspect several of them.
- Run agglomerative clustering with single, complete and Ward linkage. Plot the three dendrograms side by side and comment on how differently they partition the same data.
- Stability test: bootstrap the data 50 times, recluster, and measure how often pairs of samples land together. Report the mean pairwise co-assignment for your chosen k, and for k plus one.
- Now the important part. Add a fake "batch" variable that is correlated with two of the features, recluster, and show that you can manufacture a convincing cluster structure that has no biological content.
- Write the two sentences you would need in a paper to justify claiming these clusters are biologically meaningful, and say honestly whether your analysis supports them.

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Self-check

- Silhouette scores on standardised WDBC: 0.343 at k=2, 0.314 at k=3, 0.283 at k=4, 0.158 at k=5. What is the defensible reading?
- A) There are exactly two subtypes
- B) Two clusters is the best-supported partition among those tested, though the moderate score indicates substantial overlap
- C) The clustering has failed
- D) More clusters would score better

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Self-check

- Which clustering method can decline to assign a point to any cluster?
- A) k-means
- B) Agglomerative clustering
- C) DBSCAN
- D) None of them

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References and further reading

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Questions

Next week: continue to Week 10.

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